

Database Design — Final Schema

- Seven core tables: ParkingLot, ParkingSlot, Vehicle, RateCard, ParkingSession, Payment and Staff.
- Primary and foreign keys connect the entities and preserve referential integrity.
- RateCard is linked to ParkingLot so pricing can vary by lot and vehicle type.

Relationships

- One ParkingLot has many ParkingSlot records, RateCard records and Staff members.
- One Vehicle can have many ParkingSession records; each session uses one slot.
- Each completed session generates one Payment; a staff member can process many payments.

Stored Procedures and Triggers

- `sp_allocate_slot()` registers the vehicle if required and finds the first free matching slot.
- `sp_process_exit()` calculates duration, applies the grace period and generates the bill using the lot-specific rate.
- Triggers automatically change slot status to OCCUPIED on session creation and FREE when a session is completed.

Views and Reporting

- vw_lot_occupancy shows total, free, occupied slots and occupancy percentage.
- vw_active_sessions shows currently parked vehicles and live parking duration.
- vw_daily_revenue summarizes transactions and revenue by date.

Query Demonstration

- Master-data SELECT statements verify lots, slots, vehicles, rate cards and staff.
- Procedure calls demonstrate allocation and billing.
- The exact query assigned during Presentation-II must be inserted in the marked query-solution file before final upload.